

Config-Driven RL for Just-in-Time Adaptive Interventions

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Problem Definition

Goal: Optimise personalised physical-activity interventions using RL.

MDP formulation: $\mathcal{M} = (\mathcal{S}, \mathcal{A}, \mathcal{P}, \mathcal{R}, \gamma)$

- $\mathcal{S}$: 36-state factored MDP (step bin, sleep quality, day of week, burden)
- $\mathcal{A}$: 4 actions (idle, movement suggestion, goal reminder, journal)
- $\mathcal{P}$: Unknown — estimated via LLM simulation or hand-curated tables
- $\mathcal{R}$: $\alpha \cdot f(\text{step_bin}) + (1-\alpha) \cdot g(\text{sleep}) - \lambda \cdot \mathbb{I}[\text{action} \neq \text{idle}]$

Core challenge: How do we iterate on RL agents without live patients?

Approach: Build a config-driven simulation framework with LLM-bootstrapped user models.

Existing Work

	HeartSteps V2	PEARL	StepCountJITAI
Type	MRT	4-arm RCT	Simulator
Sample	$n \approx 97$	$n = 13,463$	Synthetic
Algorithm	TS bandit	RL bandit	DQN
Actions	~ 5 types	155 nudges	4 types
Outcome	78.4% improved	+296 steps*	DQN ≈ 3000
Theory	SDT / FBM	COM-B	—

*RL vs control at 1 month,

$$p = 0.0002.$$

Key gap: No existing framework combines *config-driven experiment definition* with *LLM-bootstrapped user simulation* for rapid agent prototyping before real-world deployment.

Our Approach

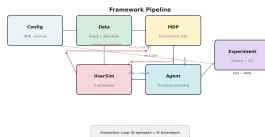
Two pillars:

① Config-Driven Framework

- Entire MDP defined in YAML — no source code changes
- 36-state factored environment, expression-based rewards, 3 transition types
- 7 RL/bandit agents, evaluation runner with multi-seed benchmarking

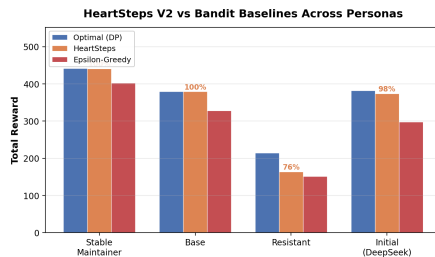
② Synthetic LLM Patients

- 22,320 prompts per persona, 5 distinct behavioural personas
- LLM-as-transition-model: DeepSeek V4 Flash samples next-state probabilities
- Total API cost: **\$0.27** for the entire bootstrap



Initial Results

HeartSteps V2 (Liao et al., 2019) reproduced with Bayesian linear regression, dosage tracking, and proxy value function.



Persona	Optimal DP	HeartSteps (% of opt)	Best Bandit (% of opt)
Stable Maintainer	441.8	440.8 (99.8%)	402.5 (91.1%)
Base	379.7	379.1 (99.9%)	327.8 (86.3%)
Resistant	214.6	163.5 (76.2%)	151.7 (70.7%)

HeartSteps achieves 97–100% of optimal across all 7 persona configurations.

What is Next

① Sensor Foundation Models

- Wearable FM POC (NormWear, 768-dim embeddings) — feature branch
- FM-guided dynamic action spaces (15 interventions, 3 archetypes)

② Additional LLM Backends

- Beyond DeepSeek V4 Flash: GLM 5.2, Qwen, Claude
- Multi-turn persona dialogues for richer state transitions

③ Prompt Engineering

- COM-B behaviour-change taxonomy integration
- Structured JSON schema enforcement, chain-of-thought prompting

④ Synthetic Clinical Results

- Calibrate transition tables to real-world effect sizes (PEARL: +200–300 steps)
- Integration with wearable datasets (All of Us, UK Biobank)